Recording

EN 580.133
Other RNA-based therapeutics

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Created by Elaine Zhelan Chen

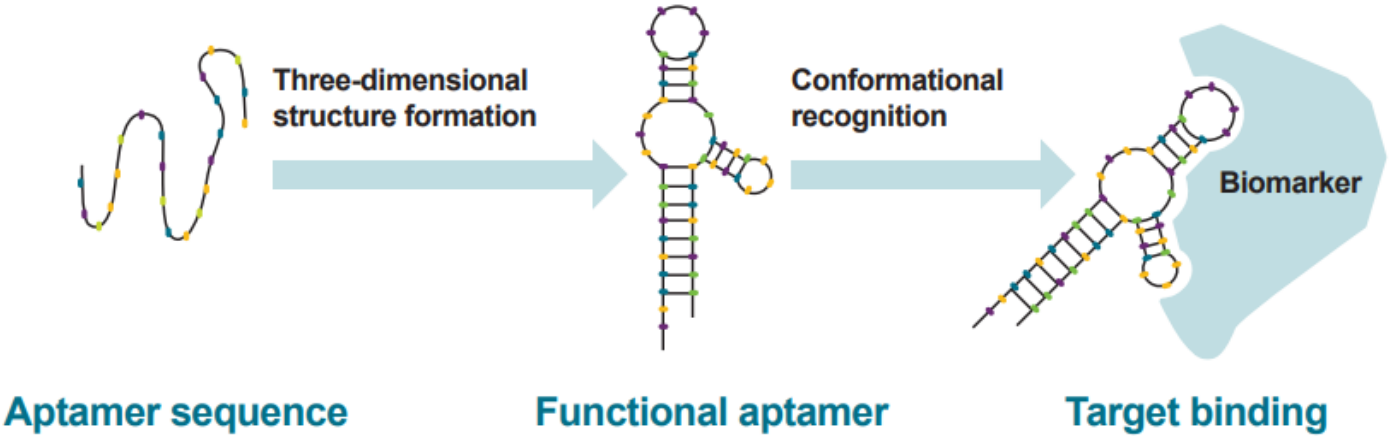
Recap

- Challenges for ASO drugs
- ASO chemical modifications
  - Phosphorothioate (PS), 2’-MOE, 2’-F, 2’-OMe, LNAs, PMO
- Approved ASO drugs
  - Case #1 hATTR, RNaseH1 mediated mRNA cleavage
  - Case #2 DMD, alternative splicing, restore reading frame
  - Case #3 SMA, alternative splicing, copy gene
  - Case #4 Batten disease, alternative splicing, N-of-1 tailored ASO

Aptamers as therapeutics

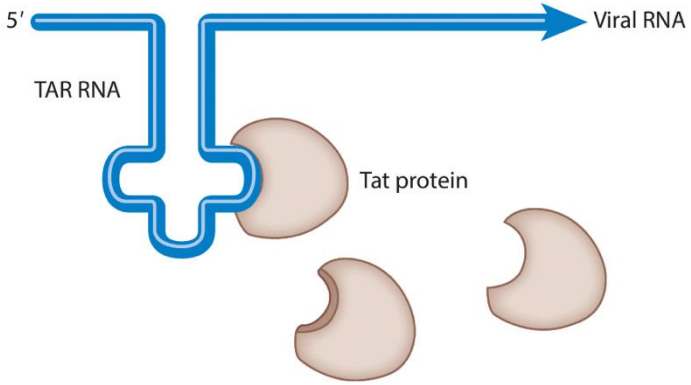
Aptamer

- Single-stranded DNA or RNA molecules
- Aptamer binds to protein targets by folding into a three-dimensional conformation



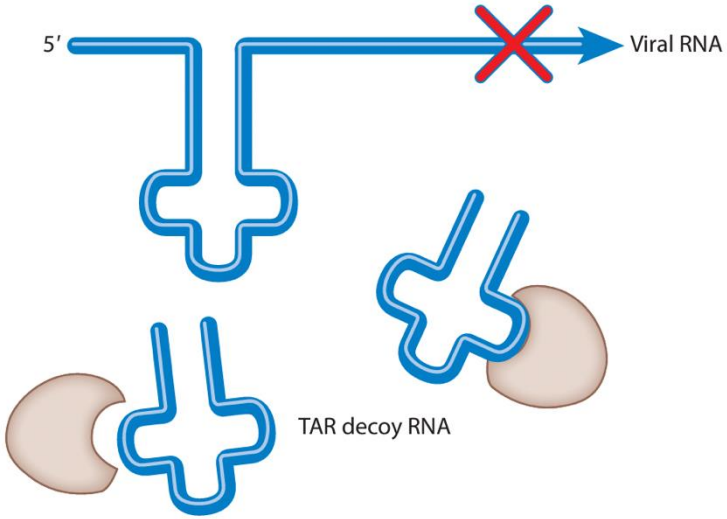
Aptamer discovery

- Discovered in human immunodeficiency virus (HIV) and adenovirus
- These viruses encoded small, structured RNA called trans-activation response or TAR
- Viral protein Tat binds to TAR, which lead to strong increase in the viral transcripts
- This facilitates viral replication or mitigated antiviral activity by the host



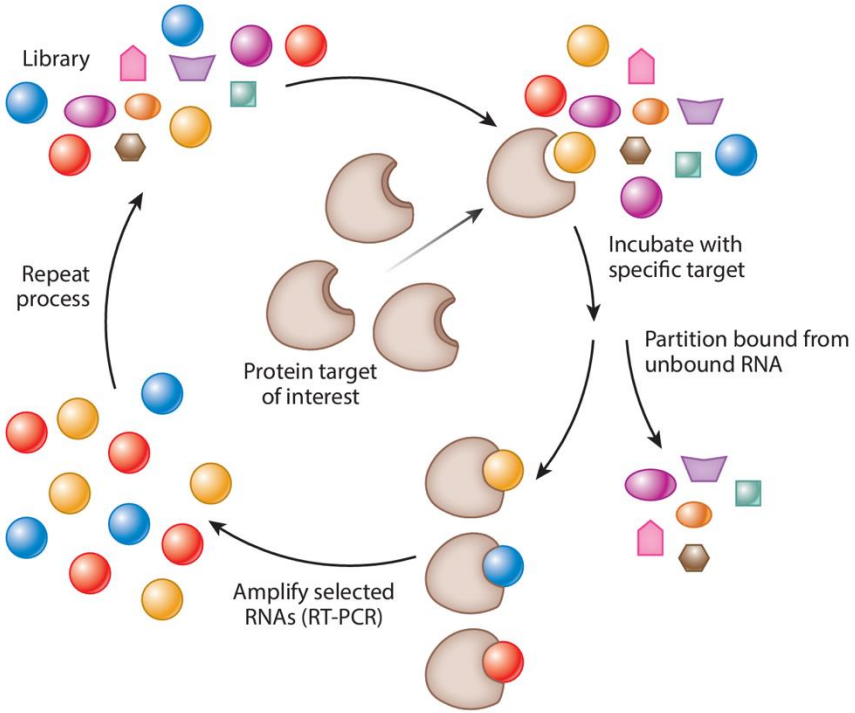
Aptamer for therapeutic purpose proof of concept

- A TAR RNA–derived aptamer was expressed in CD4+ T cells
- This aptamer served as a decoy and bind to the Tat protein
- Cells that expressed the TAR aptamer were made resistant to viral replication
- Initial demonstration that RNA aptamer can potentially serve as therapeutic agents



SELEX

- A method to screen a large library of RNA molecules to find ligands that bound to a target of interest with high affinity
- Starts with a large library of randomized nucleic acid sequences
- Each RNA molecule has a randomized region containing around 40 nucleotides
- Theoretically, there are 4^{40} different sequences, approximately 10^{15} different molecules
- The screening process is repeated 8–12 times until an RNA pool with high affinity for the target protein is isolated



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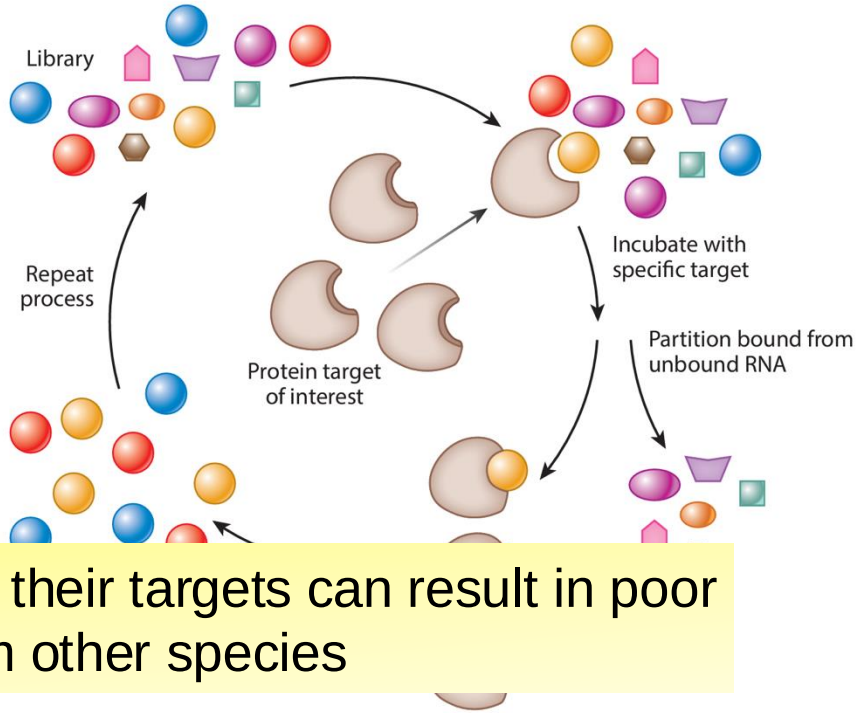
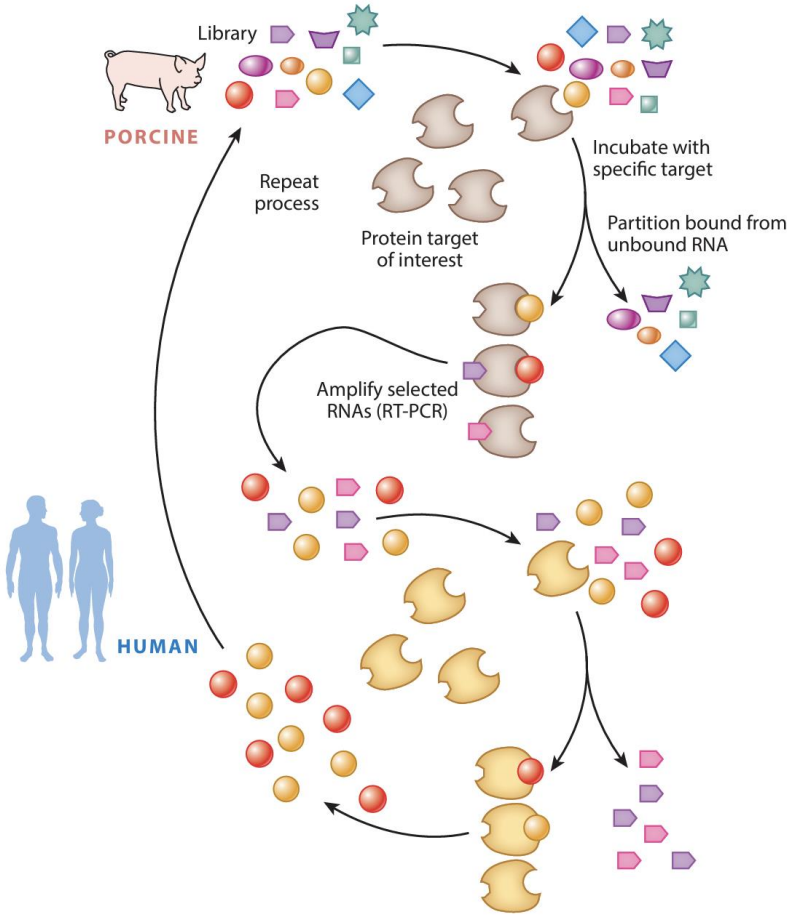
- High specificity binding of aptamers to their targets can result in poor or no cross-reactivity to homologs from other species



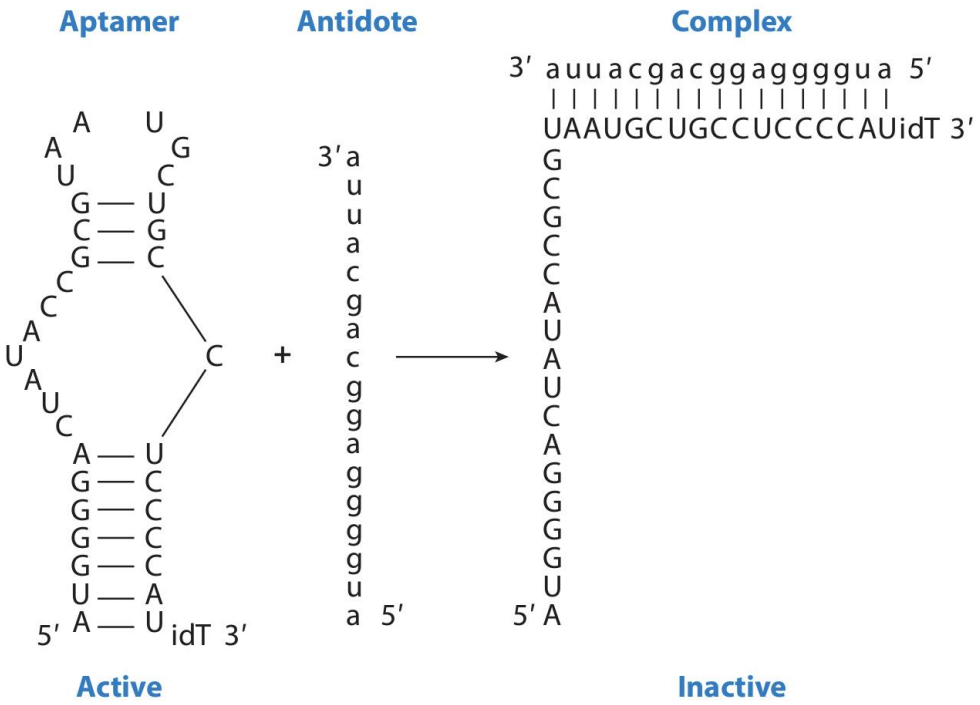
Toggle SELEX

- A modified method based on SELEX
- The RNA pool is exposed to human protein target to select candidates with high affinity
- The RNA pool with high affinity for the target protein is isolated and then exposed to the animal ortholog of the protein of interest
- This iterative exposure, first to human then to preclinical species, continues sequentially
- In the end, the aptamers that bind to both the human and animal protein are identified



Antidote

- Antidote oligonucleotides bind to the aptamers by Watson-Crick base pairing
- They can prevent the aptamers from interacting with its target protein
- In preclinical studies, the oligonucleotide antidote to coagulation factor IXa demonstrated complete reversal less than 5 min after administration
- The effect of aptamers can be reversed without relying on the liver and the kidney



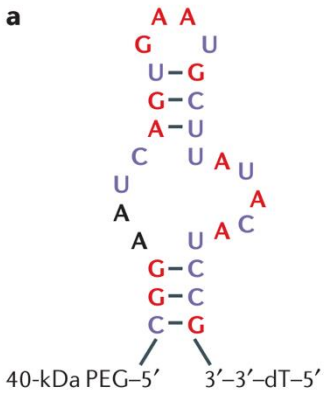
Aptamer vs Antibody

	Aptamer	Antibody
Production	Chemical synthesis	Produced in engineered cell line
Stability	Return to original conformation after temperature insult	Temperature sensitive and undergo irreversible denaturation
Size	Smaller in size, improved transport and tissue penetration compared to antibodies	
Batch consistency	Uniform activity regardless of batch synthesis	Activity of antibodies varies from batch to batch

Approved Aptamer - Pegaptanib

- Pegaptanib (Macugen)
- Developed by Eyetech Pharmaceuticals and Pfizer
- Approved in 2004

- For treating age-related macular degeneration (AMD)
- Characterized by a deterioration of the central part of the retina
- AMD is the leading cause of severe and irreversible vision loss in the elderly in the developed world

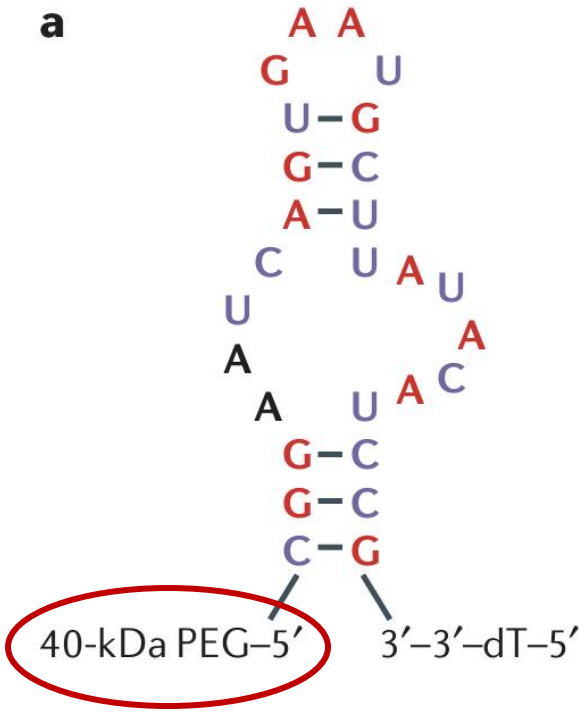


Approved Aptamer - Pegaptanib

- Pegaptanib is a 28-nucleotide RNA aptamer
- Binds specifically to extracellular VEGF
- Inhibits the binding of VEGF to its receptors and suppresses pathological angiogenesis

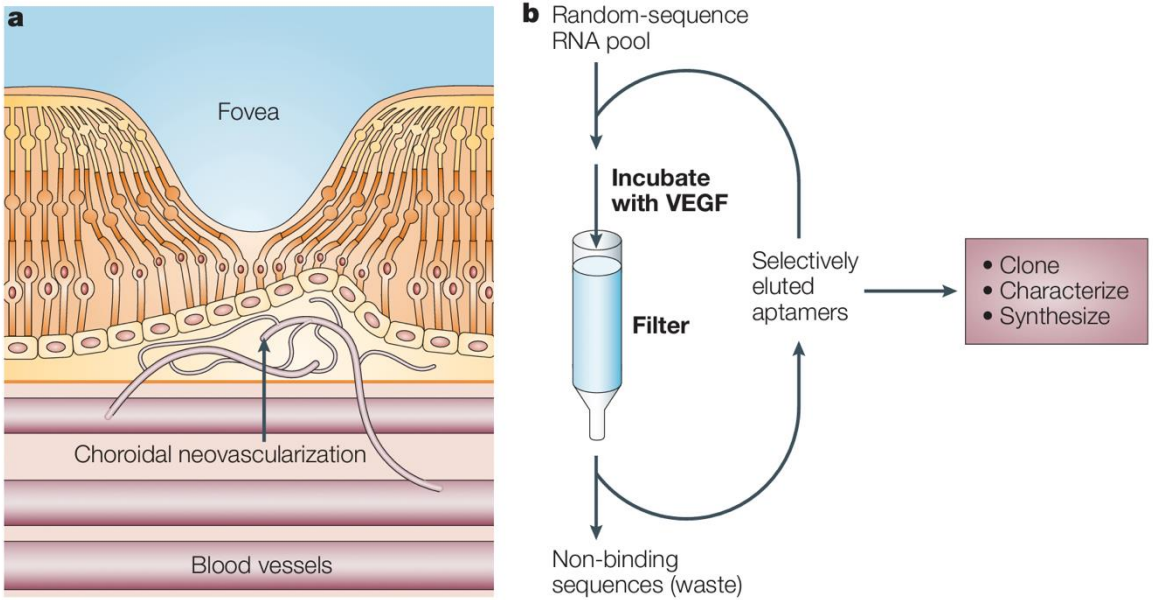
- The sugar backbone of pegaptanib is modified to prevent degradation by endogenous endonucleases and exonucleases

- Pegaptanib is conjugated to a 40-kD PEG molecule at the 5' end
- PEG conjugation
  - Increases circulation time in the bloodstream
  - Prevent rapid clearance by the kidneys



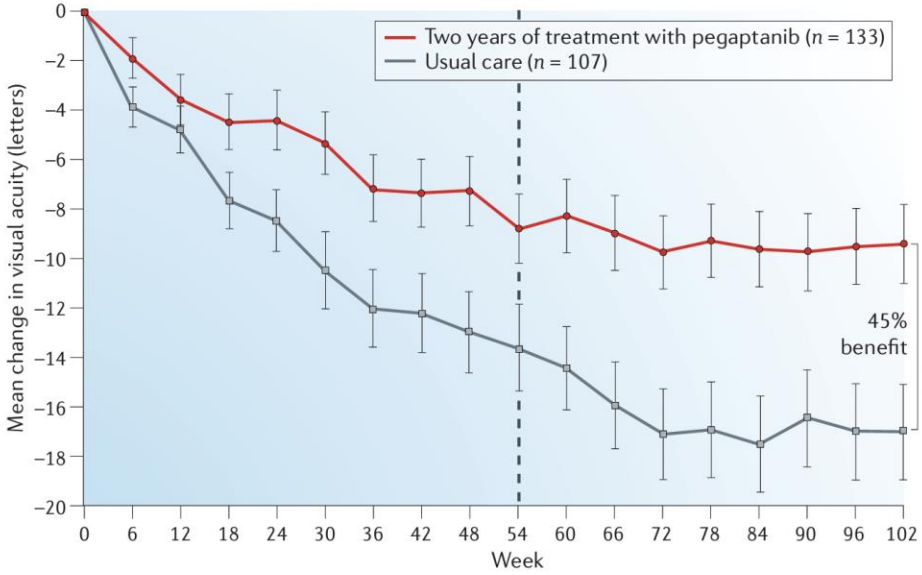
Approved Aptamer - Pegaptanib

- AMD with neovascularization, leading to leakage of blood or serum and detachment of the retinal pigment epithelium
- At the time, there were no well-established effective therapeutic interventions appropriate for all forms of neovascular AMD



Approved Aptamer - Pegaptanib

- The aptamer was administered every 6 weeks intravitreally
- Slower decline in visual acuity over a 2-year period with pegaptanib
- The beneficial effect of pegaptanib was seen as early as 6 weeks after treatment compared to negative controls
- Effective with even the lowest dose of 0.3mg
- First aptamer-based drug to receive approval from the FDA



Approved Aptamer - Izervay

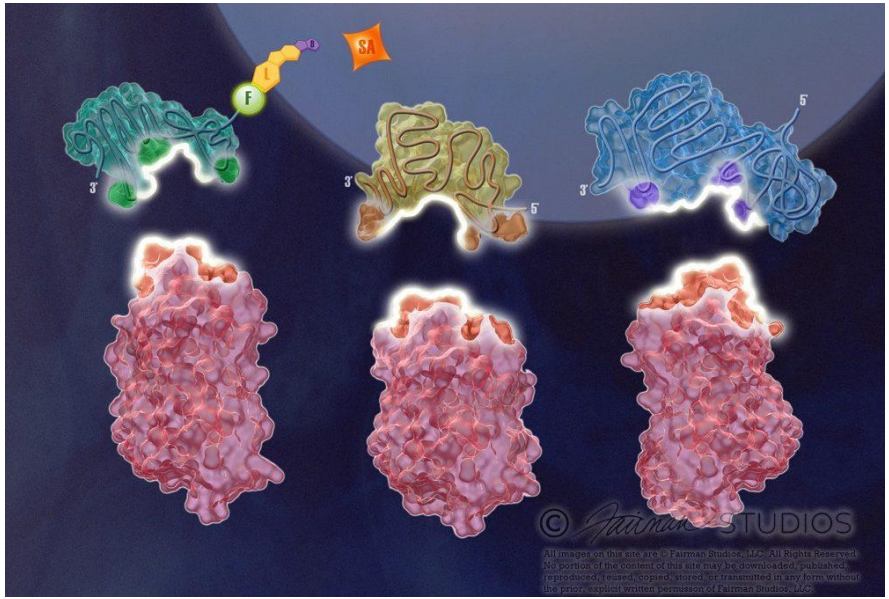
- Developed by Iveric Bio
- Approved in 2023

- For treating geographic atrophy (GA)
  - a severe form of AMD
- C5 inhibitor
  - Slow the progression of retinal cell degeneration
  - Slow the development of GA by about 20%

Aptamer – other applications

- Diagnostics
  - SOMAmers
  - Molecular probes or biosensors to disease biomarkers
  - Highly multiplexed
  - Characterize over 1,100 proteins in a small amount of protein samples

- Drug delivery
  - Conjugated to LNP for mRNA drug targeted delivery
  - Conjugated to nanocarriers for anti-cancer drugs for targeted delivery to the cancerous cells

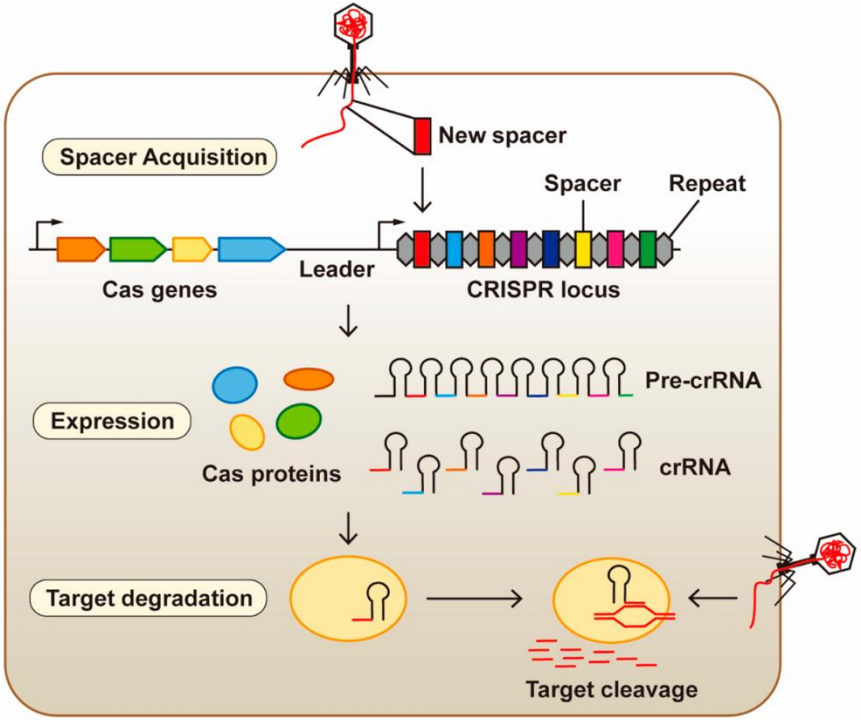


CRISPR/Cas9 gene editing

CRISPR/Cas9 discovery

- Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)
- Discovered in archaea by Francisco Mojica

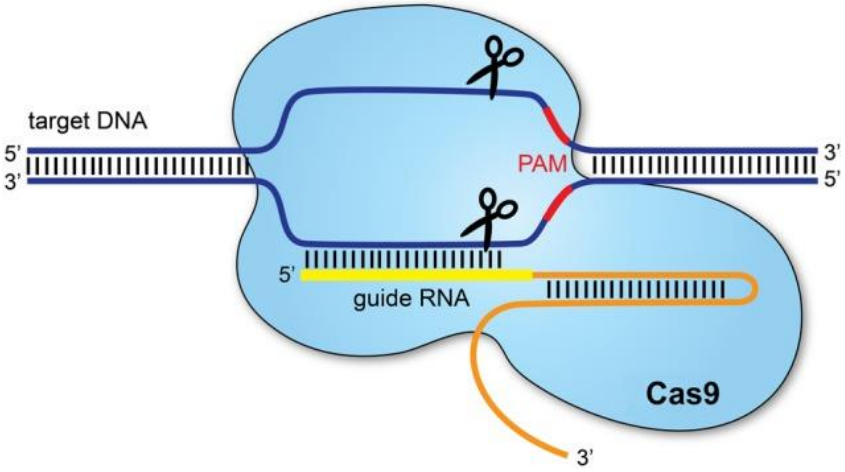
- A bacterial defense system
- Spacer sequences acquired from past invaders
- Spacer sequences are transcribed into short RNA sequences, known as crRNA
- When the target sequence is found, Cas9 binds to the DNA and make a double-stranded cut



CRISPR/Cas9 system

- CRISPR/Cas9 is a gene-editing technology
- There are two essential components:
  - a guide RNA to match a desired target gene
  - Cas9 (CRISPR-associated protein 9), an endonuclease which causes a double-stranded DNA break

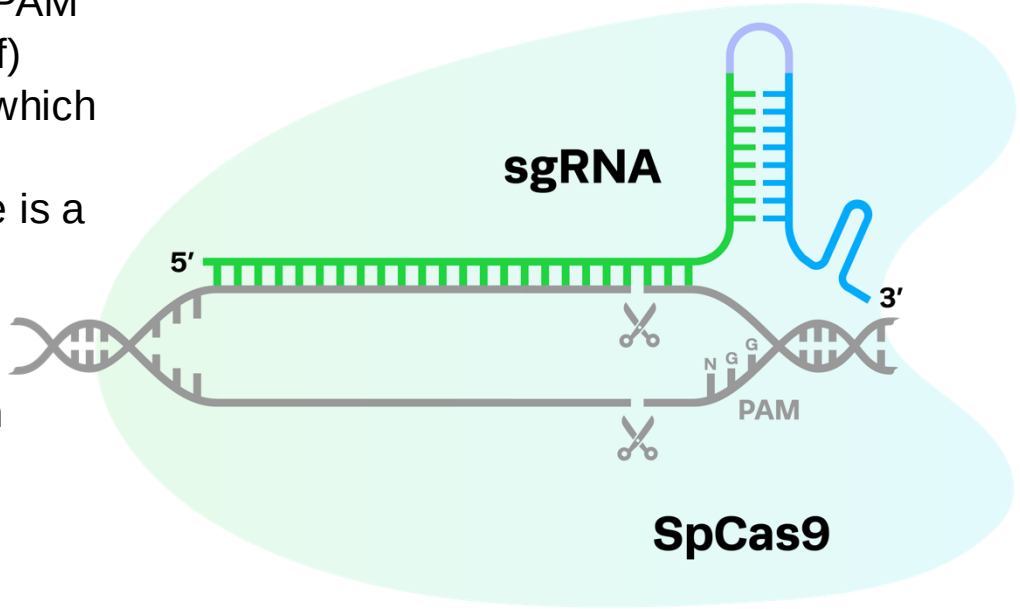
- Can be programmed to target specific stretches of genetic code and to edit DNA at precise locations



CRISPR/Cas9 mechanism

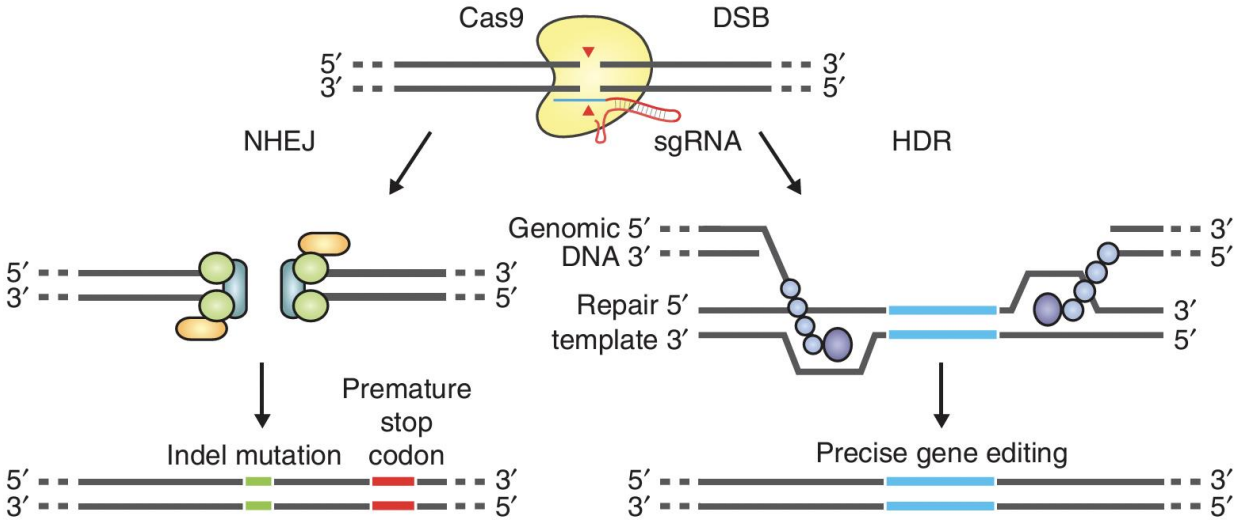
- Cas9 enzyme makes a double-stranded cut on the DNA 3 bases upstream of the PAM sequence (protospacer-adjacent motif)
- The PAM sequence is dependent on which Cas enzyme is used
- For Cas9 enzyme, the PAM sequence is a 5’-NGG-3’ sequence
    - N can be A, T, C, or G

- **At this point, no mutation has been introduced!**



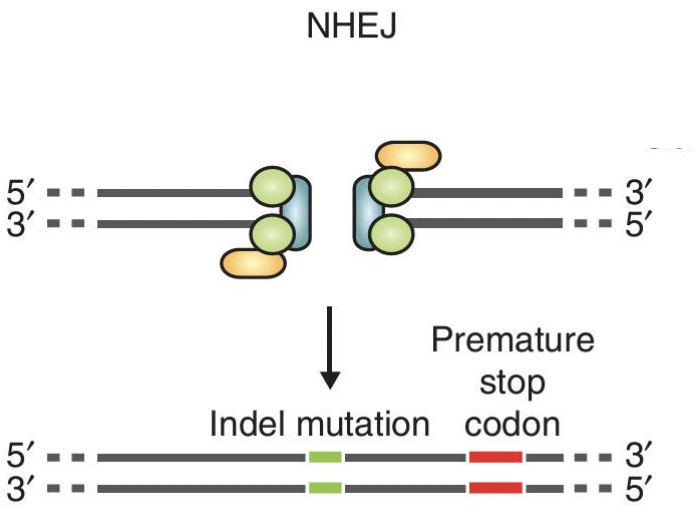
CRISPR/Cas mechanism

- Cell's DNA repair mechanism repair the dsDNA break through NHEJ or HDR
  - NHEJ: non-homologous end joining
  - HDR: homology-directed repair



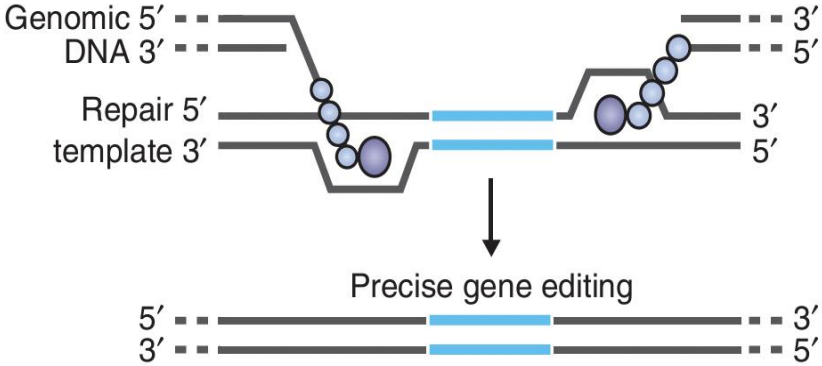
Non-homologous end joining (NHEJ)

- Repair by NHEJ prevalent, happen in about 75% of the case
- NHEJ is error-prone, and it usually results in insertions and deletions (indels) in the region being repaired
- Indels at the coding region of a gene can result in a frameshift mutation
- Functional copies of the protein encoded by the gene cannot be made as a result
- CRISPR can be used to generate knockout (KO)
- All phases of the cell cycle



Homology-directed repair (HDR)

- A donor template is provided as a DNA template for the repair to achieve a knock-in (KI)
- CRISPR knock-in can be used in cell and gene therapies to correct genetic mutations that cause human disease
- Lower efficiency than NHEJ
- Late S phase to G2 phase



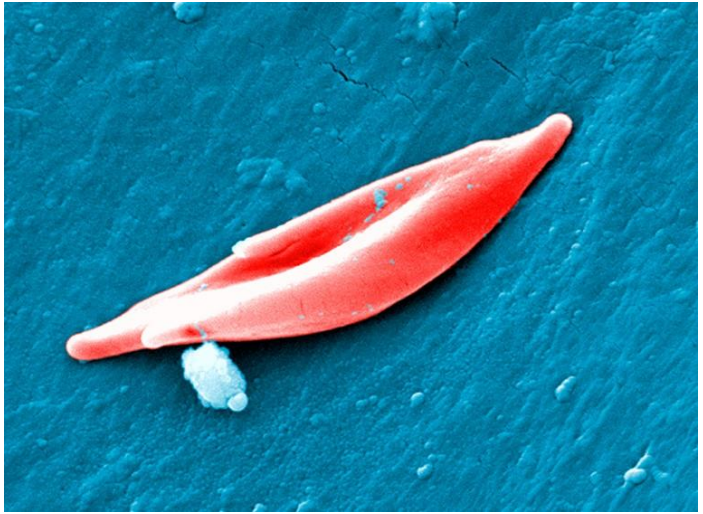
A method for genome editing

- Nobel Prize in Chemistry 2020
- Emmanuelle Charpentier and Jennifer A. Doudna

World’s first CRISPR therapy

- Casgevy
- Developed by Vertex Pharmaceuticals and CRISPR Therapeutics
- Approved in Dec 2023

- For treating Transfusion-dependent β-thalassemia (TDT) and sickle cell disease (SCD)
- The most common monogenic diseases worldwide, annual diagnosis in approximately
    - 60,000 patients with TDT
    - 300,000 patients with SCD



Transfusion-dependent β-thalassemia (TDT) and sickle cell disease (SCD)

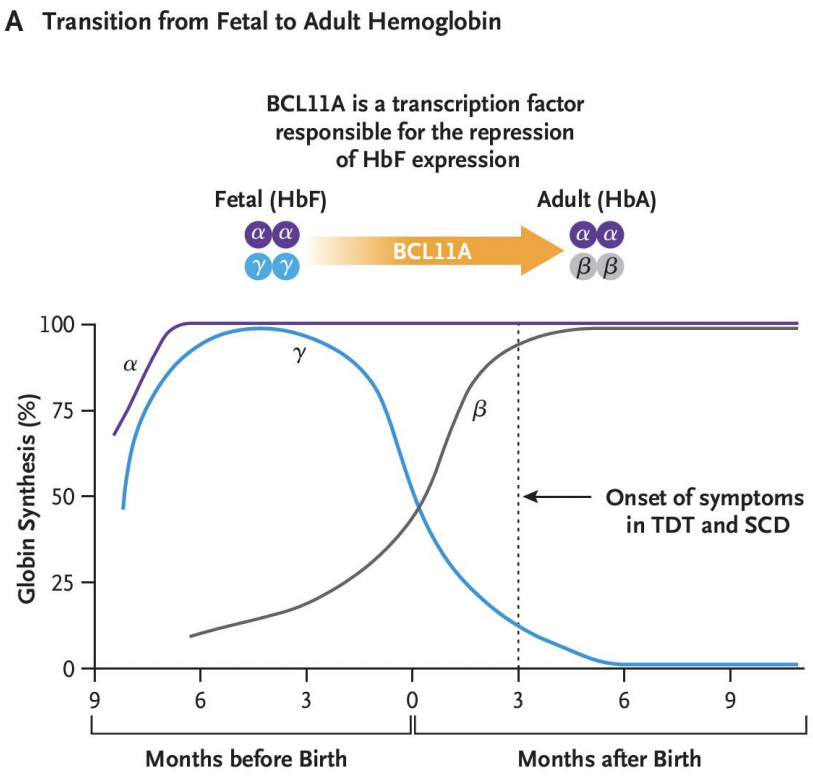
- Both are caused by mutations in the hemoglobin β subunit gene (HBB)

- For TDT
  - Mutation led to ineffective erythropoiesis
  - Need regular transfusion of red blood cells to survive

- For SCD
  - Mutation causes erythrocyte deformation, hemolysis, anemia, painful vaso-occlusive episodes, irreversible end-organ damage, and a reduced life expectancy
  - Need transfusion and pain management

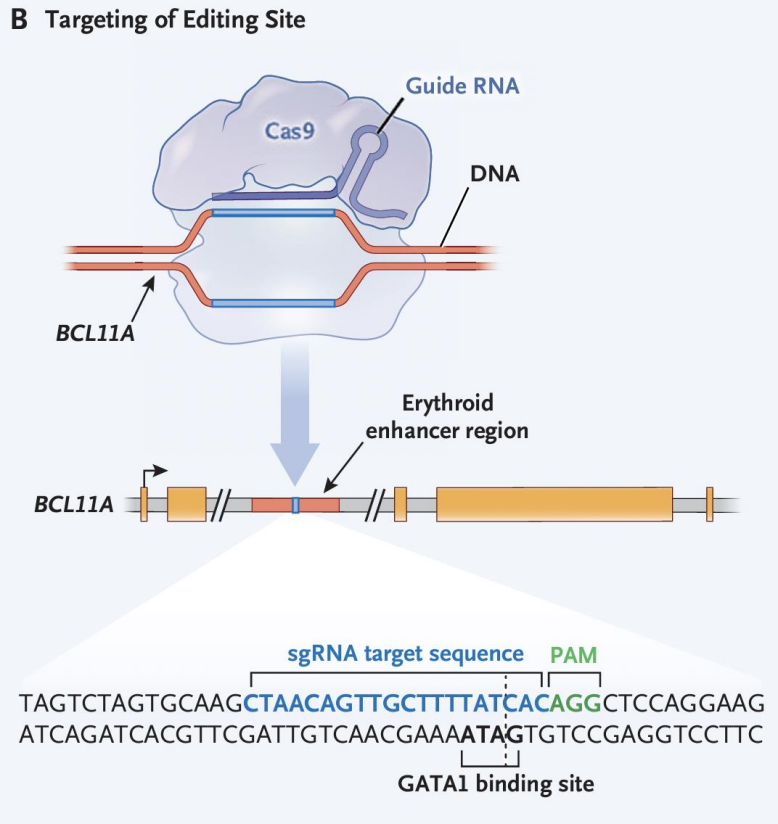
Casgevy mode of action

- Neonates and infants with TDT or SCD are typically asymptomatic while their fetal hemoglobin levels remain high
- Patients with TDT or SCD who co-inherit hereditary persistence of fetal hemoglobin, in which fetal expression continues into adulthood, have little or no disease



Casgevy mode of action

- CRISPR-Cas9 gene-editing in hematopoietic stem and progenitor cells (HSPCs) at the erythroid-specific enhancer region of BCL11A to reduce BCL11A expression in erythroid-lineage cells, restore γ-globin synthesis, and reactivate production of fetal hemoglobin



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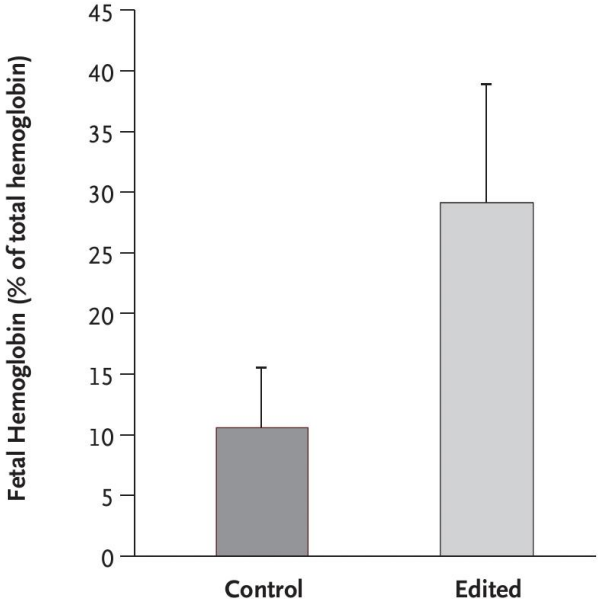
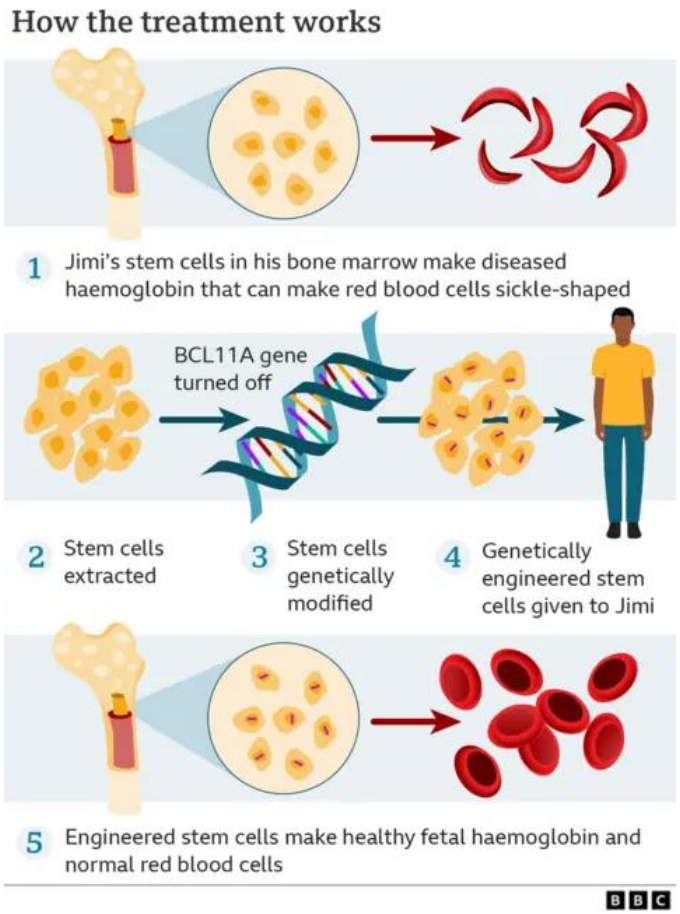


Casgevy mode of action

- CD34+ hematopoietic stem and progenitor cells obtained from the patient are electroporated
  - Electric pulse creates temporary pores within the cell membrane
  - Enables entry of the CRISPR/Cas9 complex into the cells
- Entire gene-editing process takes approximately 6 days
- The edited cells undergo quality control and are infused back into the patient

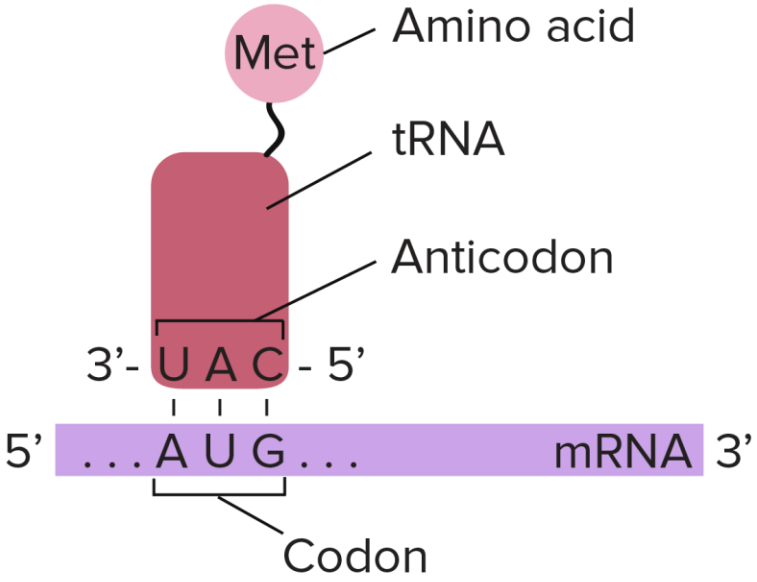


tRNA as therapeutics

tRNA

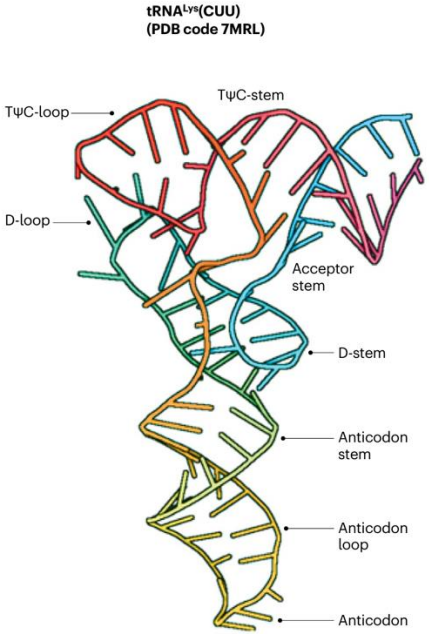
- Highly abundant cytoplasmic RNA in eukaryotic cells
- Plays a critical role in decoding mRNA and synthesizing protein

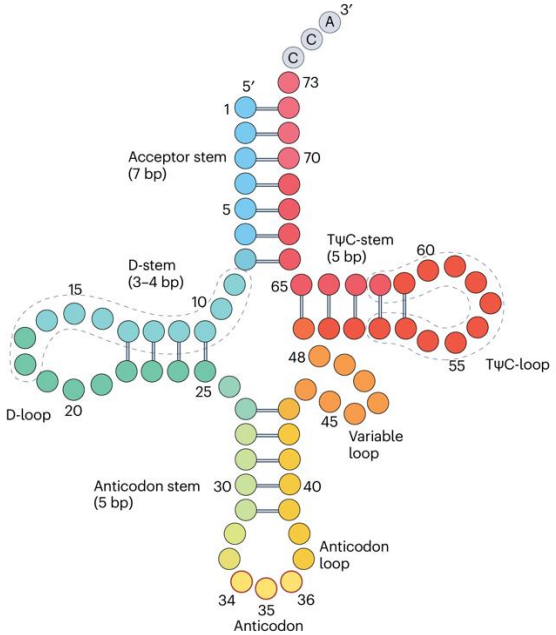
- Anticodon, a set of three nucleotides, can bind to one or a few specific mRNA codons



tRNA structure

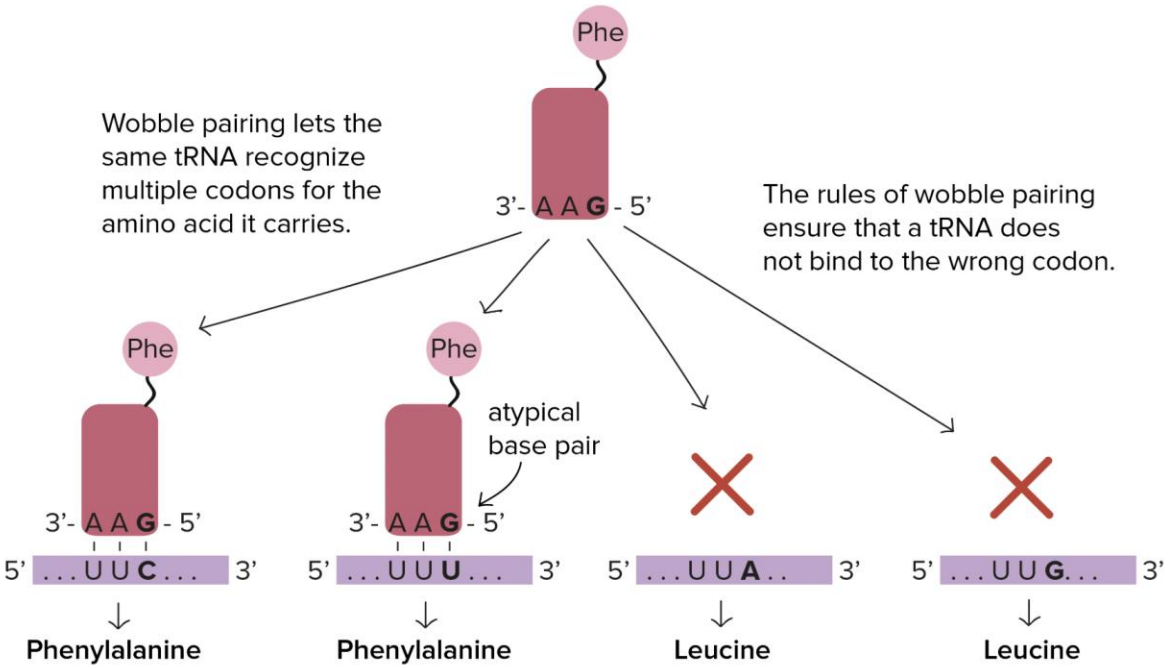
- Made from a single strand of RNA
- tRNA length range from 73 to 90 nt
- However, the anticodon is always numerated 34 to 36
- Base pairing between nucleotides in different parts of the molecule makes double-stranded regions and loops, folding the tRNA into an L shape



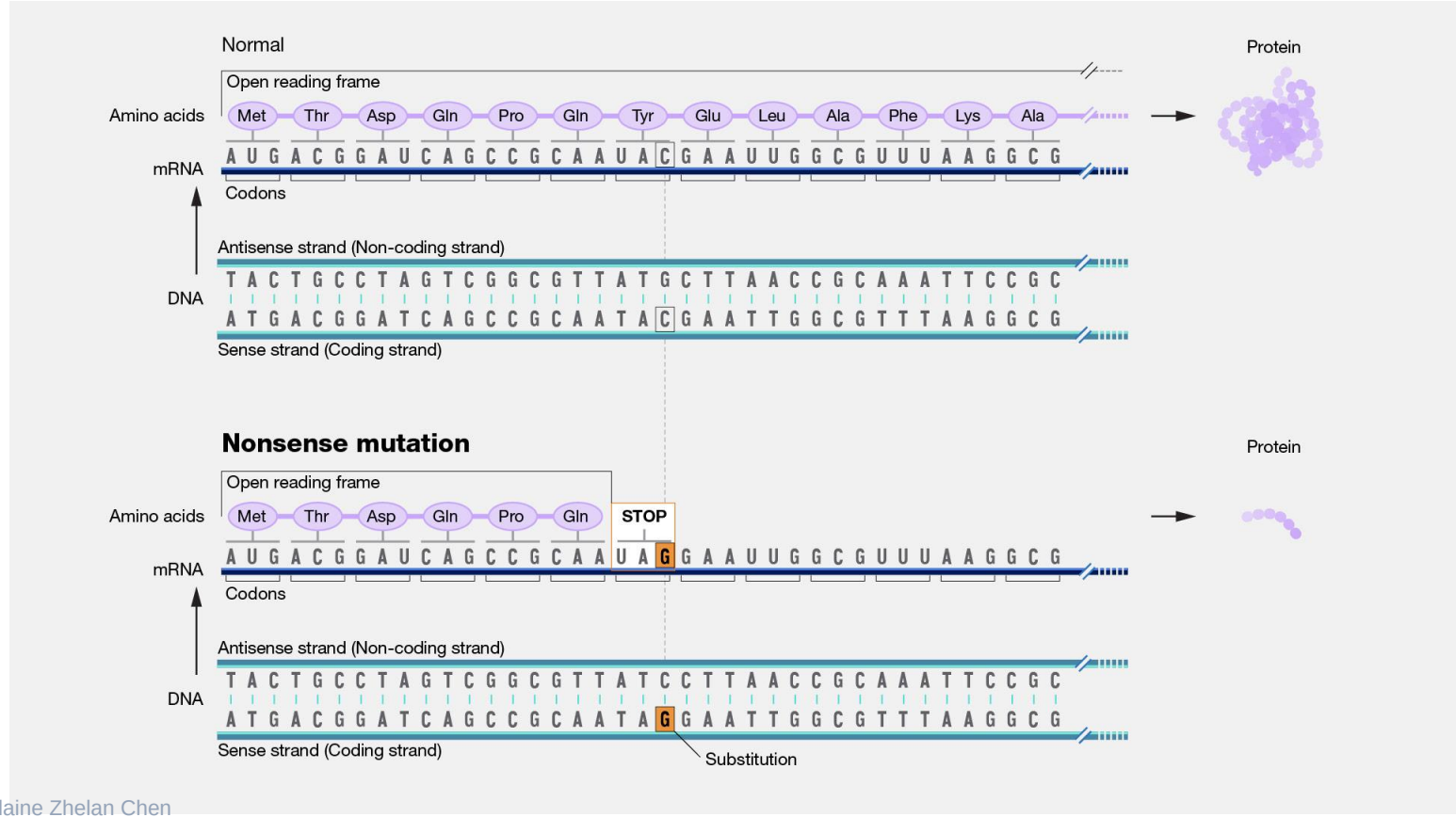


tRNA wobble pairing

- tRNAs undergo extensive modifications with an average of 13 modifications per tRNA in mammals
- Modifications at position 34 enable wobbling base pairing
- Less tRNA species, about 35–55, are necessary to decode all 61 codons

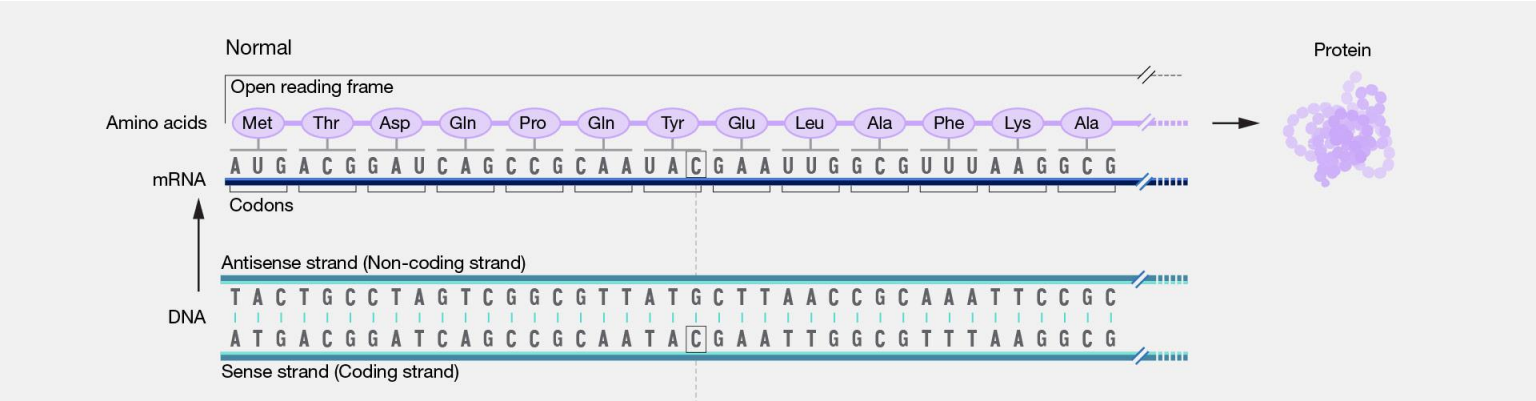


Nonsense mutation

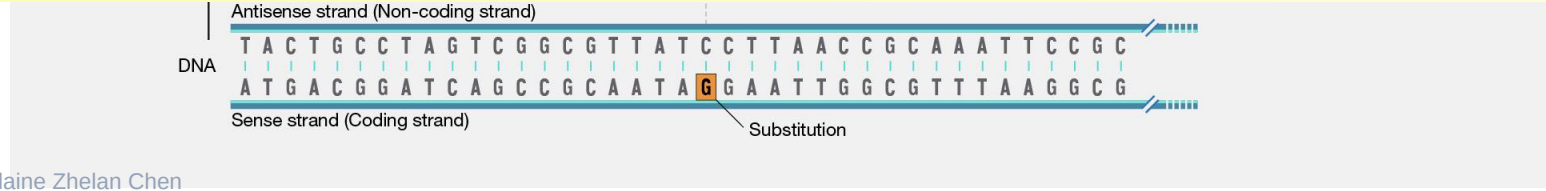


Created by Elaine Zhelan Chen

Nonsense mutation



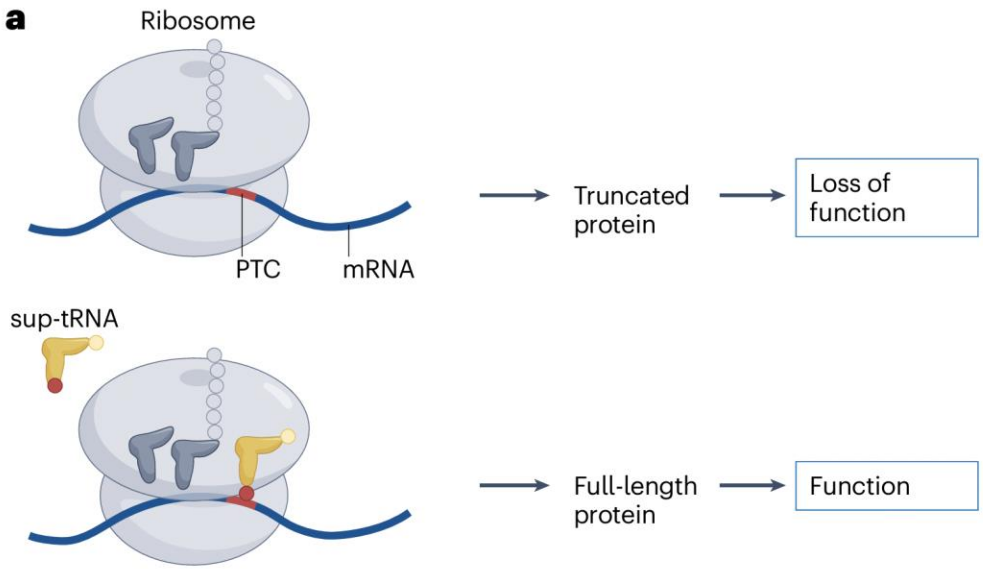
- Nonsense mutation leads to a pretermination codon (PTC), which results in a loss of protein function
- Nonsense mutations account for approximately 11% of genetic diseases



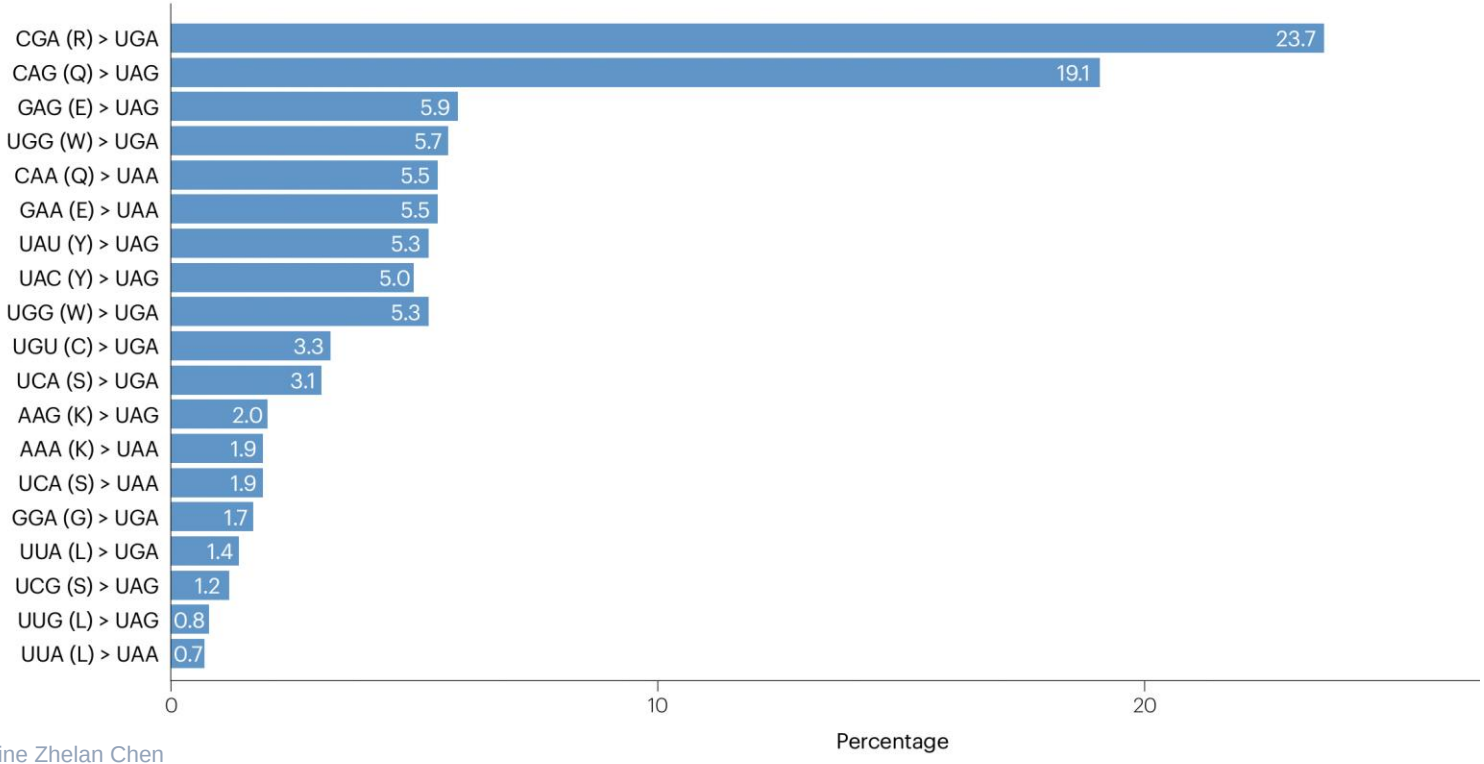
tRNA as therapeutics

- sup-tRNAs (natural suppressor tRNAs)
- sup-tRNAs are found in bacteria and yeast which can translate a stop codon

- sup-tRNAs decodes the premature stop codon with an amino acid which meant to be coded can be supplied to the cells
- sup-tRNAs can partially restore the translation of the entire mRNA sequence (readthrough) and produce functional proteins



Frequency of nonsense mutations in the human



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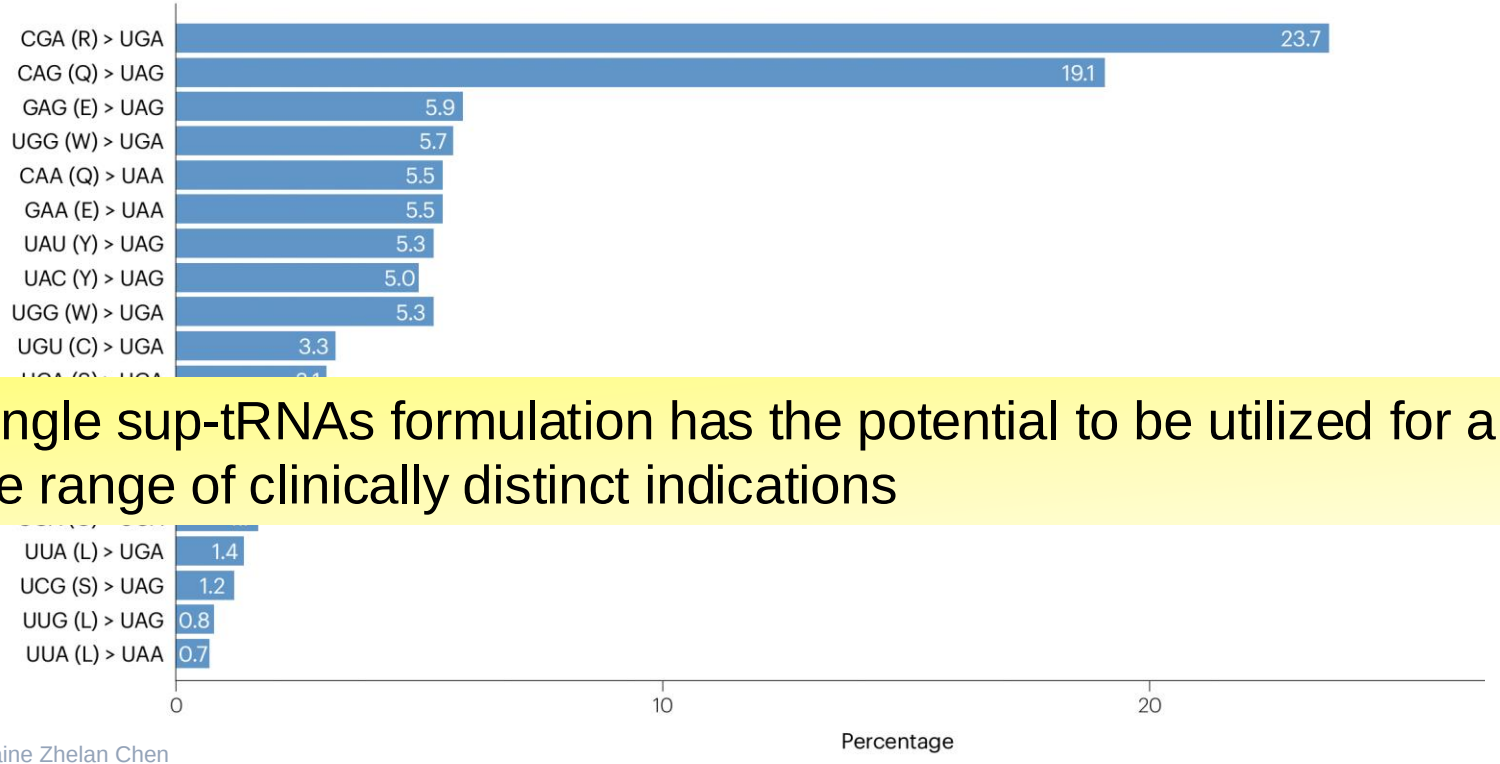


A single sup-tRNAs formulation has the potential to be utilized for a wide range of clinically distinct indications

Alltrna – tRNA solution for stop codon diseases

JOHNS HOPKINS MEDICINE

Liver

TARGET	DELIVERY	PHASE			
Arg-TGA	LNP	Discovery	Preclinical	IND-Enabling	Clinical

Arginine to TGA (Arg-TGA) liver stop codon disease can affect patients with Arg-TGA mutations in any genetic disease of the liver, including phenylketonuria, organic acidemias, and urea cycle disorders. The Arg-TGA mutation is the most prevalent nonsense mutation in the liver.

VIEW LESS

TARGET	DELIVERY	PHASE			
Gln-TAG	LNP	Discovery	Preclinical	IND-Enabling	Clinical

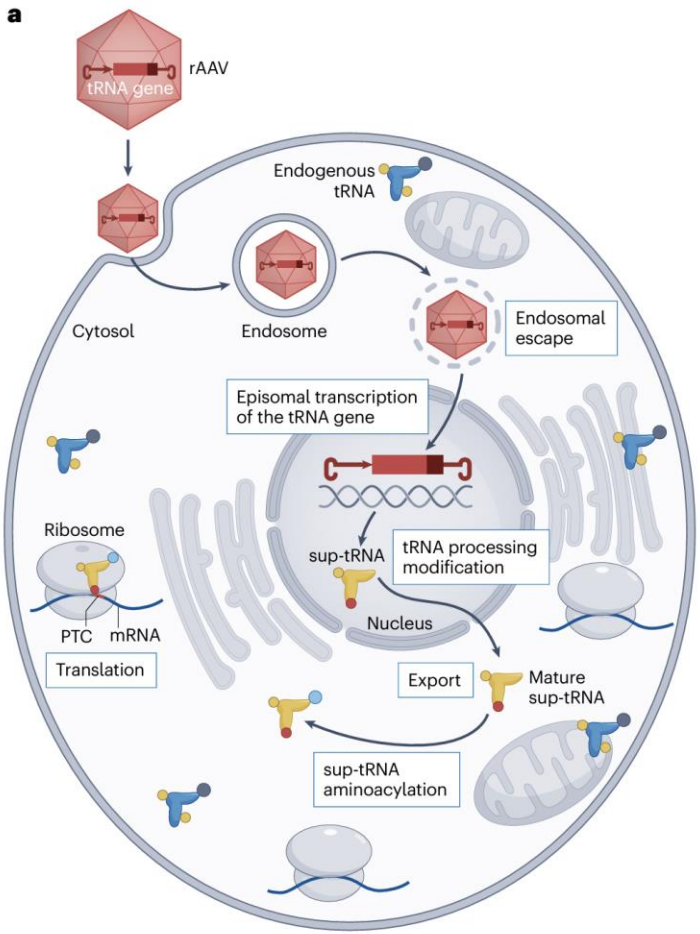
VIEW MORE

TARGET	DELIVERY	PHASE			
Other nonsense mutations	LNP	Discovery	Preclinical	IND-Enabling	Clinical

VIEW MORE

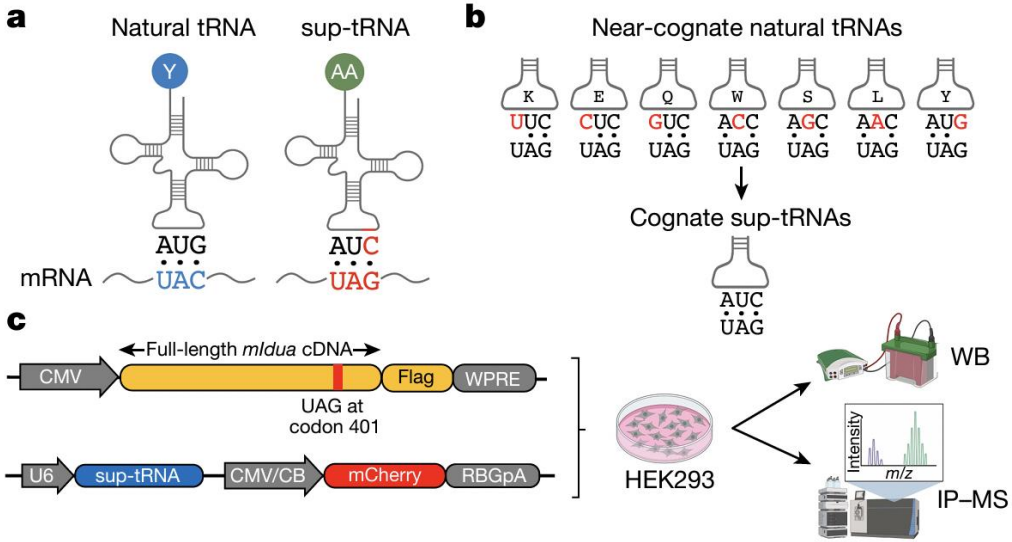
tRNA delivery - rAAV

- Recombinant adeno-associated virus (rAAV)
  - Clinically approved viral vector-based delivery vehicle
- Delivers a tRNA gene-expressing cassette to the nucleus
- Utilizes the endogenous transcription machinery, RNA polymerase (Pol) III
- Enables continuous expression of the tRNA in the target tissue with a single administration



tRNA Preclinical study - 1 AAV

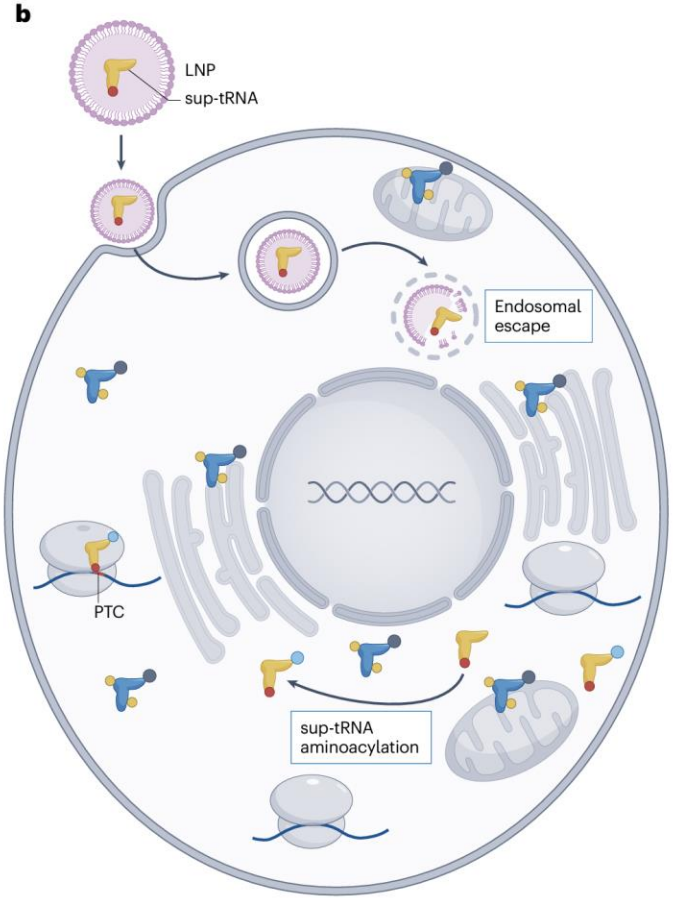
- rAAV delivery of a sup-tRNAs safely and efficiently rescued a genetic disease in a mouse model carrying a nonsense mutation
- The effects lasted for more than 6 months after a single treatment
- sup-tRNAs had a limited effect on global readthrough at normal stop codons
- sup-tRNAs did not perturb endogenous tRNA homeostasis



tRNA delivery - LNP

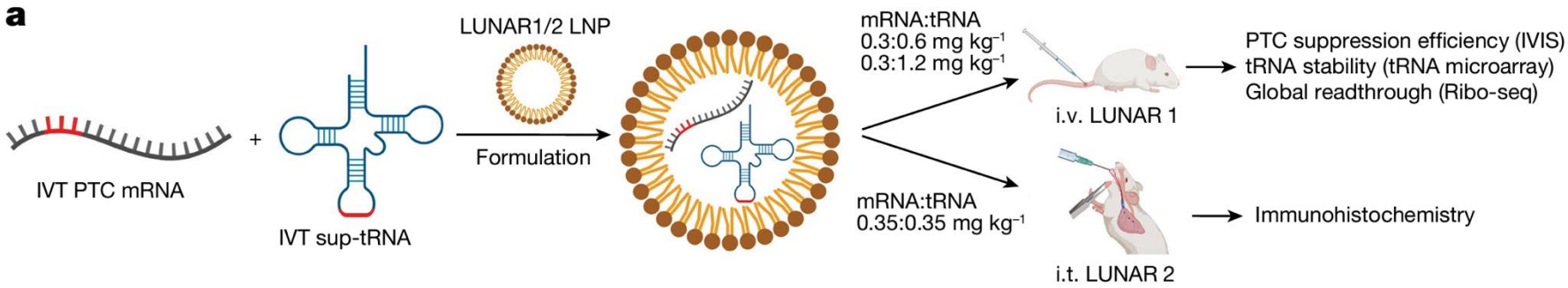
- Supply operational tRNA directly in the cytosol
- Therapeutic tRNA is readily available for translation
- tRNA can be synthesized in using in vitro transcription and/or chemical synthesis

- Studies have shown robust and persistent suppression of PTCs in mouse liver using LNP



tRNA Preclinical study - 2 LNP

- Intravenous and intratracheal LNP administration of sup-tRNA in mice restored the production of functional proteins with nonsense mutations
- sup-tRNA formulations caused no discernible readthrough at endogenous native stop codons
- sup-tRNA re-established expression and function in CFTR patient-derived nasal epithelia



tRNA challenges

- Reading through natural stop codons
- Inducing changes of tRNA repertoires may unlock cell reprogramming into a more proliferative, potentially cancerous state
- tRNA manufacture and purification need further development
- Lack of model system for rare to extremely rare diseases
- Small patient population, may need to conduct basket trials
    - Multiple diseases that have common molecular alternations

Wrap up

RNA therapeutic – One to watch

- Common challenges
  - Degradation
  - Targeted delivery and extrahepatic delivery
  - Off-target effects
  - Innate immune stimulation
  - Better understanding of long-term performance

Thank you!

Reminders

- **Submit teammate evaluation Today, Jan 16th, by 5PM EST**

Attendance